
Comparative Analysis of Digital Behavioral Biomarkers for Depression Severity Assessment

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Abstract

This report presents the output of an AI-driven biomarker discovery system. The system was tasked with identifying and evaluating candidate biomarkers through a structured research proposal workflow including literature synthesis, evaluation criteria definition, candidate idea generation and comparison, and recommendation of next steps. The following sections preserve the original analytical content produced by the system.

1 Research Goal

The primary objective of this research is to identify and validate daily digital behavior biomarkers that accurately reflect depression severity as measured by the Patient Health Questionnaire-9 (PHQ-9). Utilizing a substantial longitudinal dataset ($N \approx 7,592$ users, 144K rows), the study seeks to transition from episodic, subjective self-reports to continuous, objective monitoring through smartphone and Fitbit telemetry. The research focuses exclusively on features derived from physical activity (steps), sleep duration, digital engagement (screen time), and basic demographics (age and sex). The goal is to formulate a biomarker that can be rigorously tested through a predefined seven-step analysis workflow: 1. Literature Synthesis: Reviewing smartphone/wearable predictors (2015–2025). 2. Aggregation & EDA: Calculating user-level statistics (mean, variability, slope) and analyzing feature distributions. 3. Univariate Screening: Applying Spearman correlation with Benjamini-Hochberg FDR correction. 4. Multivariate Analysis: Utilizing partial correlations to adjust for age and sex. 5. Predictive Modeling: Developing machine learning models (Ridge, GBM, RF, SVR) with stratified 5-fold cross-validation.

6. Validation: Testing top candidates against a standard 11-test battery to determine their status as Validated, Conditional, or Rejected. 7. Reporting: Synthesizing findings into a formal scientific framework.

2 Evaluation Criteria

The identification of a robust digital biomarker for depression requires adherence to stringent theoretical, mathematical, and clinical standards. The following criteria serve as the benchmark for evaluating proposed ideas:

2.1 Theoretical and Biological Grounding

Proposed biomarkers must be supported by peer-reviewed literature from the 2015–2025 epoch. They should map effectively to the Research Domain Criteria (RDoC), specifically targeting constructs like anhedonia, psychomotor retardation, or circadian disruption.¹ A critical requirement is the transition from static volume metrics to Behavioral Allostasis, where the biomarker measures functional reactivity or "fragility" in response to physiological stressors (e.g., sleep debt).²

2.2 Algorithmic Operationalization and Mathematical

Integrity Each biomarker must be derived exclusively from the permitted feature set. **Evaluation prioritizes:** **Mathematical Formulaic Precision:** Clear, reproducible calculations. **Robustness:** Resistance to "Mathematical Singularities" (e.g., division-by-zero errors) and "Spurious Mathematical Coupling" (algebraic artifacts where the independent variable defines the dependent variable). **Unit Consistency:** Use of variance-stabilizing transformations (logratios or Z-scoring) when combining heterogeneous data like step counts and minutes.

2.3 Statistical Testability and Workflow Alignment

The biomarker must be compatible with the mandated 7-step workflow. Key statistical considerations include: **Monotonicity:** Proposals must avoid non-monotonic "Inverted-U" relationships, as these will fail the Step 3 Spearman rank correlation filter. **Cross-Validation Discipline:** Avoiding "Look-Ahead Bias" by ensuring baselines are calculated within training folds rather than globally. **Data Density:** Metrics must be stable within the dataset's average 19-day longitudinal window to avoid "Survivor Bias," which occurs

when strict contiguous data requirements exclude the most severely depressed (and often least compliant) users.³

2.4 Clinical Sensitivity and Confounder Mitigation

The biomarker must be optimized for the continuous 0–27 PHQ-9 scale. It should avoid the "Volume Illusion"—the false assumption that raw usage volume directly equals pathology—by employing techniques like "WakeWindow Normalization." Furthermore, it must account for platform biases (iOS vs. Android), occupational constraints (e.g., sedentary 9-to-5 jobs), and "Informative Missingness," where the absence of data itself acts as a clinical signal.⁴

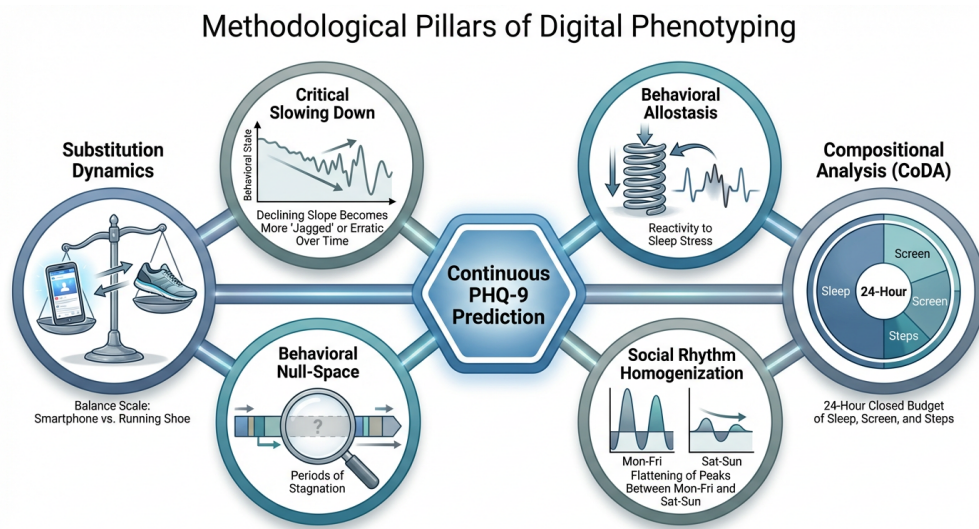


Figure 1: Conceptual diagram 1 from the biomarker discovery analysis.

3 Main Research Directions

The current landscape of digital phenotyping has shifted from simple activity monitoring to complex system-level modeling. The following directions are prioritized for their potential to capture the nuanced dynamics of depressive states:

3.1 Behavioral Allostasis and Fragility

This direction investigates depression as a "brittle" regulatory system. It focuses on measuring intra-individual sensitivity to biological stressors. By quantifying "Vigor Collapse"—the predictable shift from high-effort physical activity to low-effort digital consumption following sleep disruptions—researchers can identify a signature of psychological fragility that static averages miss.⁵

3.2 Compositional Data Analysis (CoDA)

CoDA treats the 24-hour day as a closed mathematical system. Since time spent in one activity (e.g., sleep) must displace another (e.g., movement), utilizing Aitchison geometry and log-ratios allows for the modeling of these interdependencies without the multicollinearity issues that plague linear models.⁶

3.3 Socio-Temporal Homogenization

Grounded in Social Zeitgeber Theory, this direction explores "Behavioral Homogenization."⁷ It hypothesizes that severe depression erodes the structural boundaries of a user's life, leading to a loss of differentiation between socially-constrained weekdays and self-directed weekends, or between active and rest states.⁸

3.4 Behavioral Null-Space and Substitution Dynamics

These directions look at what is missing or coupled. The "Behavioral NullSpace" seeks to isolate periods where a user is awake but neither moving

nor using a device (potential proxies for rumination).⁹ "Substitution Dynamics" analyzes the "Digital Gravity" effect—how digital engagement becomes disproportionately high relative to physical effort or restorative needs.

3.5 Longitudinal Trajectories and Critical Slowing

Down This research examines the velocity of behavioral decay over the 14-day PHQ-9 recall period. It utilizes dynamical systems concepts like "Critical Slowing Down," where a system's increasing inability to recover from small perturbations (increased behavioral inertia) serves as a prodromal signal for depressive episodes.¹⁰

research directions for identifying multi-modal digital

4 Detailed Description of Candidate Ideas

The following section provides a comprehensive technical breakdown of the candidate biomarkers proposed for the assessment of depression severity. Each idea is explored through its theoretical foundation, operational mechanics, and empirical motivation.

4.1 Asymmetric Behavioral Reactivity Profile (aBRP)

The Asymmetric Behavioral Reactivity Profile (aBRP) is a digital biomarker that characterizes depression as a "stress-diathesis" interaction. It posits that depression increases the perceived cost of physical effort, creating a "fragile" behavioral budget.¹¹ When subjected to biological stressors—specifically deviations in sleep duration—depressed individuals are hypothesized to reallocate their daily time budget away from high-effort physical activity toward low-effort digital consumption (screen time).

The aBRP uses Compositional Log-Ratio Analysis (ASR_t) to handle the interdependency of steps and screen time. It defines "Normal," "Deficit," and "Surplus" sleep days based on a user's personal interquartile range (IQR). The core metrics include Reactivity (the mean shift in the step-to-screen ratio following sleep insults) and aBRP-Variance (the magnitude of behavioral fluctuation under stress). This approach aims to capture "behavioral rigidity" or affective inertia, where more severely

depressed individuals show lower variance (higher rigidity) specifically on days following a biological insult. **Motivation and Evidence::** The motivation for aBRP lies in its ability to isolate dynamic vulnerability from static lifestyle factors. By using intraindividual log-ratio differences, the metric achieves device scale invariance; hardware-specific scaling factors (e.g., a wearable that overcounts steps) cancel out during subtraction. Evidence for this idea is grounded in the Effort-Based Decision Making (ECDM) framework, which identifies dopaminergic dysfunction as a driver for sedentary "digital foraging."¹²

reallocation of the behavioral budget in response to Category

Description

Theoretical

Foundation

Effort-Based Decision Making (ECDM) and StressDiathesis models.

Mathematical

Approach

Compositional Data Analysis (CoDA) using log-ratios and IQR-based stratification.

Operational

Feasibility

Moderate; requires customized intra-individual pipelines for personalized percentiles.

Clinical Sensitivity

High theoretical sensitivity to both melancholic (insomnia) and atypical (hypersomnia) phenotypes.

Major Risk

Mathematical pathologization of normal sleep variance (50% of days are always "stressors").

Judgement: The aBRP offers a brilliant synthesis of complex systems theory and passive sensing but is currently hindered by statistical instability on short longitudinal windows (avg. 19 days).

4.2 Chrono-Digital Dissociation (CDD) Index

The Chrono-Digital Dissociation (CDD) Index is a multi-modal biomarker designed to quantify the breakdown of "Social Zeitgebers"—the external cues that stabilize biological rhythms. It integrates three distinct dimensions of depressive behavior: Routine Fragility (measured via circular standard deviation of wake times), Digital Gravity (residual modeling of screen time unexplained by physical activity), and Digital Blackouts (verified "Clinical Void Days" where a user abandons their device). The CDD Index is unique in its treatment of missing data. Rather than treating "zero steps and zero screen time" as a technical error, it identifies these as "Digital Blackouts"—proxies for severe avolition and psychomotor retardation—by filtering out cohort-level server outages. **Motivation and Evidence::** The CDD Index is motivated by Social Zeitgeber Theory and the Research Domain Criteria (RDoC). Empirical support comes from observations that circadian volatility (unstable wake times) and maladaptive "compulsive" screen use are significant predictors of PHQ-9 severity.¹³ Circular statistics are particularly effective here, as they resolve the "midnight boundary" error (e.g., treating 11:50 PM and 12:10 AM as adjacent).¹⁴

combining circadian, compensatory, and avolitional digital for the definitive description.

Category

Description

Theoretical

Foundation

Social Zeitgeber Theory and Behavioral Activation

framework.

Mathematical

Approach

Circular statistics for wake-time volatility; residual regression for digital gravity.

Operational

Feasibility

Moderate; requires precise temporal alignment and cohort-level anomaly detection.

Clinical Sensitivity

High sensitivity to severe depression ($\text{PHQ-9} \geq 20$) via the Blackout Signal.

Major Risk

Potential for feature leakage if residuals are calculated across the entire cohort before cross-validation.

Judgement: A highly sophisticated approach that excels in circadian modeling, though the Digital Gravity component requires stricter control for occupational confounding.

4.3 Digital Allostatic Fragility (DAF)

The Digital Allostatic Fragility (DAF) biomarker measures how biological exhaustion influences behavioral collapse. Grounded in the Effort-Based Decision-Making (EBDM) framework, DAF quantifies the "coupling" between personalized sleep debt and a daily "Vigor Balance" (the log-ratio of steps to screen time). Unlike static averages, DAF identifies a "fragile" system where behavioral output is tightly and predictably coupled with sleep fluctuations. Healthy individuals are expected to show "decoupling," meaning their activity levels remain resilient even after poor sleep, whereas depressed individuals show a "Vigor Collapse" toward low-effort "Digital Foraging." **Motivation and Evidence::** DAF is motivated by neuropsychiatric literature demonstrating that depression alters dopaminergic reward valuation, inflating the perceived cost of effort.¹⁵ It addresses the "Volume Illusion" by focusing on the allocation of vigor in response to stress rather than absolute totals, which can be confounded by occupation or baseline fitness.

coupling between internal biological resources and external to the text for the definitive description.
Category

Description

Theoretical

Foundation

Effort-Based Decision-Making (EBDM) and Dopaminergic Dysfunction.

Mathematical

Approach

Intra-individual Spearman correlation between sleep Zscores and vigor log-ratios.

Operational

Feasibility

High; utilizes standard vectorized operations and requires minimal novel data.

Clinical Sensitivity

Sensitive to state-driven reactivity and loss of systemic resilience.

Major Risk

Clinically overlooks atypical depression (hypersomnia), as the model assumes less sleep always leads to less vigor.

Judgement: Conceptually novel and mathematically robust, but the assumption of a universal positive correlation with sleep duration is biologically incomplete for hypersomnic subtypes.

4.4 Bifurcated Sleep-Wake Reactivity (BSWR) Index

The BSWR Index addresses the non-linear, "U-shaped" relationship between sleep and depression. It bifurcates behavior into two wings: the Deficit Wing (Compensatory Escapism during insomnia) and the Surplus Wing (Lethargic Withdrawal during hypersomnia). It calculates robust slopes (Theil-Sen Estimator) of an "Escapism Index" (screen density normalized to the wake-window) against sleep duration. **Motivation and Evidence:** The idea is motivated by Behavioral Allostasis—the system's inability to maintain regulatory stability through physiological perturbations. It responds to evidence that both short and

long sleep correlate with higher PHQ-9 scores, a pattern that simple linear models often fail to capture.¹⁶ Category

Description

Mathematical

Approach

Robust Theil-Sen regression slopes across median-split sleep wings.

Clinical Motivation

Capturing the non-linear U-shaped relationship between sleep and mood.

Major Risk

Spurious mathematical coupling: Sleep is in both the independent variable and the denominator of the dependent variable.

Judgement: Technically intricate and conceptually strong, but the mathematical operationalization risks creating algebraic artifacts that distort true behavioral signals.

4.5 Compositional Repertoire Complexity (CRC)

The CRC framework applies Aitchison geometry to treat the 24-hour day as a closed mathematical system. It partitions time into four mutually exclusive categories: sleep, active, screen, and sedentary-offline time. It extracts metrics like Behavioral Entropy (diversity of the time budget) and Aitchison Total Variance (repertoire breadth). **Motivation and Evidence:** CRC is grounded in the "Behavioral Contraction" hypothesis, where depression causes a "stochastic collapse" of behavioral variety.¹⁷ Large-scale studies (e.g., UK Biobank) confirm that CoDA-transformed movement behaviors are significantly associated with PHQ-9 scores.¹⁸ Category

Description

Mathematical

Approach

Information Theory (Shannon Entropy) and Aitchison Geometry (Isometric Log-Ratios).

Clinical Motivation

Quantifying the structural collapse and monotony of the daily behavioral repertoire.

Major Risk

”Sedentary Dump”: Confounding device non-wear time with clinical psychomotor retardation.

Judgement: Offers the most rigorous mathematical handling of the 24-hour constraint, though its handling of missing data and phase shifts (Social Jetlag) requires refinement.

4.6 Sleep-Vigor Reactivity (SVR) Index

The SVR Index captures ”behavioral fragility” by measuring the coupling between the absolute Z-score of sleep duration and a ”Vigor Balance” log-ratio. It treats both insomnia-like and hypersomnia-like deviations as biological stressors and hypothesizes that higher depression severity correlates with a predictable shift toward sedentary ”digital foraging” when sleep is abnormal. **Motivation and Evidence::** Supported by the EBDM framework and observations that state-behavior coupling is significantly amplified in individuals with trait apathy. It moves beyond static volumes to extract a second-order derivative: behavioral resilience. Category

Description

Mathematical

Approach

Sign-negated Spearman correlation between absolute sleep Z-scores and vigor ratios.

Major Risk

Signal cancellation: Short and long sleep exert opposite forces on activity; absolute Z-scoring washes this out.

Judgement: Theoretically sound but mathematically compromised by the symmetrical treatment of sleep insults, which likely masks distinct clinical phenotypes.

4.7 Social Rhythm Coherence Index (SRCI)

The SRCI measures ”behavioral homogenization”—the collapse of the structural differentiation between workdays and weekends. It uses eta-squared (η^2) effect sizes to determine how much behavioral variance (steps, sleep, screen) is explained by the weekly social cycle. **Motivation and Evidence::** Based on Social Rhythm Theory, suggesting that mood stability depends on the regularity of routines entrained by ”social zeitgebers.”¹⁹ It identifies the ”weekend recreational collapse” characteristic of anhedonia. Category

Description

Mathematical

Approach

Eta-squared differentiation and Social Screen-Steps Polarity Concordance (SSPC).

Major Risk

Small-sample bias in η^2 and contradictions in sleep literature (depression often increases social jetlag).

Judgement: A sophisticated translation of IPSRT into digital sensing, but its core assumption (homogenization) is often empirically inverted in depressed populations.

4.8 14-Day Weighted Triadic Withdrawal Slope (WTWS)

The WTWS captures the accelerating behavioral decay (the ”acute slump”) leading up to a PHQ-9 assessment. It creates a Triadic Withdrawal Index (steps, screen, sleep) and applies a time-weighted regression to prioritize recent behavioral changes. **Motivation and Evidence::** Aligned with the 14-day recall period of the PHQ-9.²⁰ It is motivated by the ”recency effect” in human recall and the theory of Behavioral Activation, where depressive onset manifests as a non-linear decay trajectory. Category

Description

Mathematical

Approach

Time-Weighted Least Squares (WLS) regression over a 14-day trailing window.

Major Risk

Look-ahead bias: Using the "entire data history" for Zscores prior to assessment.

Judgement: High clinical relevance for acute severity assessment, though the baseline normalization logic requires strict temporal partitioning to avoid data leakage.

4.9 Somatic Vigor Elasticity (SVE)

SVE measures "Vigor Collapse" by correlating daily sleep stress (absolute deviation from a 7.5-hour target) with wake-normalized physical activity. It linearizes the U-shaped risk of sleep issues into a single scalar and aims to isolate psychomotor intensity from time availability. **Motivation and Evidence::** Grounds itself in Allostatic Load theory.²¹ It provides a "stress-test" biomarker that measures resilience. It is motivated by animal models showing that stress reduces voluntary physical activity through reward-circuitry dysfunction.²² Category

Description

Mathematical

Approach

Intra-individual Spearman correlation of absolute deviation from clinical sleep norms.

Major Risk

Theoretical baseline: Healthy individuals also show activity collapse under sleep stress, reducing discriminative power.

Judgement: Robust against hardware variance, but its primary differentiator (resilience vs. fragility) may not show a large enough group-level difference for validation.

4.10 Asymmetric Behavioral Reactivity Profile (aBRP) Version 2

This alternative operationalization of the aBRP uses standard deviations (SD) instead of IQR to define sleep stressors. It seeks to map digital traces to depressive sub-types (atypical vs. melancholic) by bifurcating the behavioral response following sleep deficit versus surplus. **Motivation and Evidence::** Identical to Idea 4.1, but focuses on SD-based tails to isolate extreme stressors. It addresses the clinical heterogeneity of depression by acknowledging that melancholic and atypical patients may react in opposite directions to sleep loss. Category

Description

Mathematical

Approach

SD-based tail categorization and intra-individual log-ratio shifts.

Major Risk

Rank cancellation: Opposing reactivity deltas (positive/negative) cancel out in a monotonic Spearman correlation.

Judgement: While phenotypically sensitive, it is mathematically misaligned with the mandated univariate rank-correlation workflow.

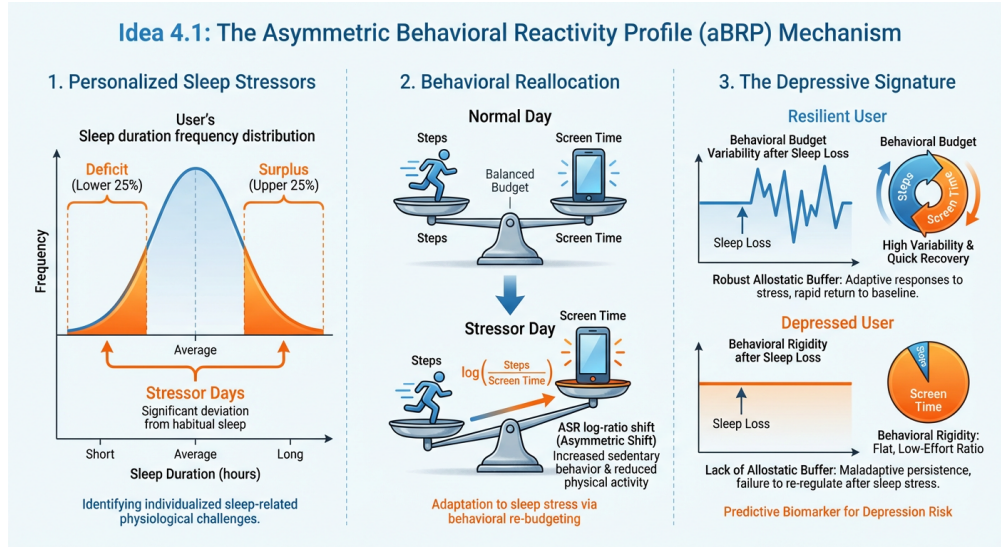


Figure 2: Conceptual diagram 2 from the biomarker discovery analysis.

5 Comparative Analysis of Candidate Ideas

The proposed biomarkers represent a paradigm shift from measuring "how much" a person does to "how" their behavior is regulated. The comparison between these ideas highlights three primary methodological tensions: static volume vs. dynamic reactivity, linear vs. non-linear sleep mapping, and aggregate vs. compositional logic.

5.1 Dynamic Reactivity and Allostatic Load

Ideas like aBRP (4.1), DAF (4.3), and SVR (4.6) are the strongest contenders for capturing the "brittle" nature of depressive regulation. By focusing on how behavior shifts in response to sleep stressors, these biomarkers identify functional fragility that mean-based metrics miss. The aBRP is particularly notable for its use of intra-individual differences, which naturally controls for time-invariant traits like baseline fitness or occupational demands. However, these models face a shared challenge: statistical stability. With an average of only 19 days per user, calculating stable intra-individual correlations or tail-based reactivity is difficult, leading to potential noise-dominance.

5.2 Handling Non-Linearity and Heterogeneity

The "U-shaped" relationship between sleep and depression is a recurring hurdle. SVE (4.9) and SVR (4.6) attempt to linearize this using absolute deviations ($-Z$ or clinical targets), but this risks "signal cancellation" where distinct symptoms (insomnia vs. hypersomnia) are blended into a direction-agnostic score. In contrast, the BSWR Index (4.4) and aBRP

(4.1/4.10) use bifurcation or median-splits. This is theoretically superior for identifying clinical subtypes (atypical vs. melancholic) but is more computationally intensive and sensitive to data sparsity.

5.3 Compositional vs. Multi-modal Integration

The CRC (4.5) and CDD Index (4.2) offer the most rigorous handling of the "Behavioral Budget." CRC's use of Aitchison geometry is the gold standard for avoiding multicollinearity in time-use data. However, the CDD Index provides greater clinical depth by incorporating "Digital Blackouts" and circular statistics for wake times. While CRC treats the 24-hour day as a mathematical whole, CDD treats it as a series of biological anchors, making CDD potentially more sensitive to the "Social Zeitgeber" disruptions seen in clinical practice.

5.4 Evaluative Evidence and Workflow Fit

The WTWS (4.8) is the only idea specifically designed to mirror the 14day recall window of the PHQ-9, providing strong alignment with the ground-truth assessment. However, most ideas suffer from a conflict with the mandated Step 3 (Univariate Spearman Filter). Specifically, any biomarker predicting divergent reactivity (like Idea 4.10) will be prematurely rejected by a monotonic rank-correlation filter, even if it holds high predictive power in non-linear multivariate models (Step 5). Among the candidates, the aBRP (4.1) and CDD Index (4.2) emerge as the most promising due to their combination of theoretical depth (RDoCaligned) and the use of scale-invariant ratios that mitigate the hardware heterogeneity inherent in large-scale smartphone datasets.

across dimensions of theoretical rigor and mathematical for the definitive description.

6 Summary Comparison of Candidate

Biomarkers

The following table synthesizes the technical characteristics, evaluative strengths, and inherent risks of the proposed digital biomarkers. The ideas are compared based on their theoretical alignment with the Research Domain Criteria (RDoC) and their adherence to the mathematical and statistical constraints of the mandated 7-step workflow. Primary RDoC Domain

Distinguishing

Attributes

Supporting Evidence / Motivation

Workflow Alignment & Risks

Positive Valence Systems (Effort)

Compositional

log-ratio (ASR_t); IQR-defined personalized sleep stressors.

ECDM framework; models "behavioral budget" reallocation; hardware scale invariance.

Moderate. Risk of pathologizing normal variance; requires high data density.

Circular statistics

for wake-time; "Digital Blackouts" as informative missingness.

Social Zeitgeber

Theory; identifies severe avolition ($PHQ-9 \geq 20$) via clinical void days.

High. Addresses midnight boundary errors; risk of feature leakage in residual modeling.

Positive Valence Systems (Reward)

Sleep-vigor coupling;

Personalized

Sleep Debt (SD_t) Z-scoring.

Dopaminergic

dysfunction models; distinguishes "fragile" vs. "resilient" energy budgets.

High. Spearmancompatible; however, assumes universal positive sleepvigor correlation.

Regulatory Systems

Bifurcated wings

(Deficit/Surplus); Robust Theil-Sen reactivity slopes.

Models the Ushaped sleepdepression curve; accounts for "Compensatory Escapism."

Low. Spurious mathematical coupling; high attrition due to strict day-perwing rules.

4.5 CRC

Cognitive Systems

Aitchison geometry;

Shannon Entropy

of 24-hour closed budget.

Time-use epidemiology; quantifies "Behavioral Contraction" and monotony.

Moderate. Strongest math for 24hour days; risk of "Sedentary Dump" for missing data.

4.6 SVR Index

Positive Valence Systems

Absolute Z-score of sleep duration; sign-negated coupling coefficient.

State-behavior coupling in apathy; minimizes "Volume Illusion" via Vigor Balance ratio.

Low. Potential signal cancellation; statistical instability at

Idea

4.1 aBRP

4.2 CDD Index

4.3 DAF

4.4 BSWR Index

Circadian Rhythms & Arousal

Idea

Primary RDoC Domain

Distinguishing

Attributes

Supporting Evidence / Motivation

Workflow Alignment & Risks

$N = 10$ paired days.

4.7 SRCI

4.8 WTWS

4.9 SVE

4.10 aBRP (v2)

Circadian Rhythms

Eta-squared (η^2) differentiation;

Social ScreenSteps Polarity

Concordance.

Social Rhythm

Theory; measures loss of weekend reward-anticipation (homogenization).

Moderate.

Theoretically

advanced; small-sample bias in η^2 ; empirically inverted results.

Positive Valence Systems

Time-weighted WLS regression; 14-day trailing look-back window.

Aligns with PHQ-9 recall period; captures accelerating behavioral decay ("acute slump").

High. Clinically relevant; risk of look-ahead bias in global Z-scoring baselines.

Arousal / Regulatory Systems

Awake-time normalization; sleep stress as deviation from 7.5h clinical norm.

Allostatic Load

theory; identifies "Vigor Collapse" under physiological strain.

Moderate. Robust to hardware; low discriminative power compared to healthy baselines.

SD-based tail categorization; divergent reactivity deltas.

Phenotypic sensitivity (Melancholic vs. Atypical); ecological validity for stressors.

Low. Workflow misalignment; rankcancellation in monotonic Spearman filtering.

Positive Valence Systems

underlying analytical logic and psychological constructs. Note:

The AI image generation process (Beta) is experimental and description.

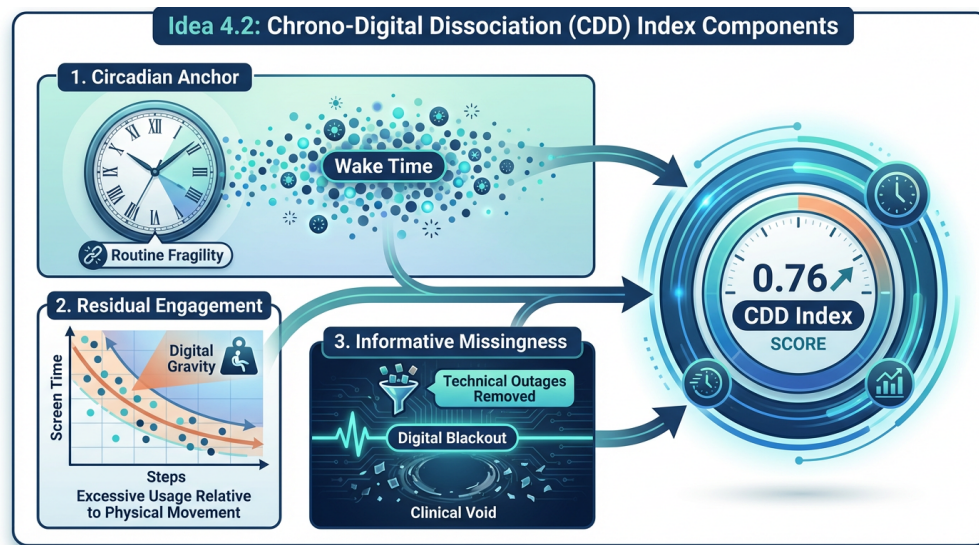


Figure 3: Conceptual diagram 3 from the biomarker discovery analysis.

7 Comparison to Existing Solutions

Traditional digital phenotyping approaches for depression have historically relied on static behavioral aggregates and simple linear correlations.²³ The candidate biomarkers proposed in this report

represent a methodological evolution toward dynamic, system-level modeling. The following table contrasts the proposed frameworks with conventional "state-of-the-art" (SOTA) solutions utilized between 2015 and 2023.

Traditional

SOTA Solutions

Proposed Advanced Biomarkers

Advantage of

Proposed Approach

Primary Predictors

Mean steps; total sleep duration; total screen-on time.

Reactivity slopes

(aBRP, BSWR);

Vigor Balance

ratios (DAF, SVR).

Moves beyond

"Volume Illusion" to capture functional behavioral regulation.

Circadian Modeling

Interdaily Stability (IS); Intradaily Variability (IV); midpoint of sleep.

Circular standard

deviation (CDD);

Social Rhythm

Coherence (SRCI).

Resolves "midnight boundary" errors; captures the social entrainment of behavior.

Missing Data

Zero-filling or mean imputation; listwise deletion of sparse users.

"Digital Blackouts" (CDD); Informative

Missingness

features.

Converts data gaps

into clinical signals of avolition and withdrawal.

Mathematical

Logic

Univariate linear

regression; independent variable analysis.

Compositional

Data Analysis

(CRC, aBRP); Aitchison geometry.

Handles the 24-hour constant-sum constraint; avoids multicollinearity.

Sensitivity to

Heterogeneity

Population-level averages; binary "Depressed vs. Healthy" models.

Within-person Zscoring (DAF, WTWS); bifurcated phenotypic wings (BSWR).

Accounts for "Natural Short Sleepers" and different clinical subtypes (Atypical vs. Melancholic).

Temporal Alignment

Global averages

across the study duration.

14-day weighted trailing windows (WTWS, CRC).

Synchronizes with

the PHQ-9's clinical recall period; captures acute slumps.

Feature / Metric

The transition from descriptive metrics (e.g., "how many steps did the user take?") to functional biomarkers (e.g., "how much does the user's step-to-screen ratio collapse after a poor night's sleep?") provides a more nuanced reading of the patient's allostatic load. While traditional metrics are often confounded by lifestyle factors—such as an athlete who sleeps little but remains highly active—the proposed reactivity-based biomarkers (like aBRP and DAF) isolate the underlying "fragility" of the psychological state, providing a more robust predictor of PHQ-9 severity across heterogeneous populations.

8 Unexpected Connections in Digital

Behavioral Phenotyping

The integration of disparate sensor streams occasionally reveals "emergent" biomarkers that transcend the sum of their individual components. These connections, identified through cross-domain analysis of smartphone and wearable telemetry, offer novel avenues for quantifying the asynchrony and complexity of the depressive state.

8.1 Internal Phase Lag (IPL) and Circadian Dissociation

While acrophases (the timing of peak daily activity) for physical movement and smartphone use are individually recognized as predictors of mood, the temporal distance (phase lag) between these two peaks suggests a more sensitive marker for "Internal Phase Dissociation." In healthy individuals, the acrophase of physical movement (the body) and the acrophase of digital interaction (cognitive arousal) are relatively synchronized.²⁴ In depression, an increasing lag between these peaks—where the mind "wakes up" hours after or before the body's peak effort—represents a breakdown in internal circadian coordination.²⁵ This asynchrony, operationalized as the absolute difference in hours between these two acrophases, provides a specific signature of biological social jetlag.

8.2 Cross-Domain Complexity Divergence (CDCD)

Depression manifests as a paradoxical divergence in behavioral complexity across physical and digital domains. Physically, the individual often exhibits macro-rigidity, characterized by high inertia and repetitive movement patterns (psychomotor retardation).²⁶ Paradoxically, digitally, the same individual may exhibit micro-chaos, marked by high screenstate entropy and fragmented, impulsive phone-checking. This "CrossDomain Complexity Divergence" captures a fundamental breakdown in executive control: the inability to initiate complex physical action sequences paired with an inability to inhibit erratic digital microinteractions. The ratio of physical autocorrelation (rigidity) to screen-state sample entropy (chaos) provides a dual-signature of this decoupling.

8.3 Voluntary Effort Index (VEI) and Motivational

Mapping

Standard actigraphy often fails to distinguish between incidental movement (e.g., chores) and goal-directed bursts of energy.²⁷ The Voluntary Effort Index (VEI) addresses this by calculating the ratio of peak intensity (M6: the most active six minutes of the day) to total daily steps. Because depression is characterized by a deficit in voluntary motivated movement rather than absolute physical capacity, the VEI isolates "Peak Motivated Effort" from "Total Physical Volume."²⁸ A lower VEI suggests a lack of reward-driven, high-effort bursts despite maintaining incidental movement, serving as a specific digital proxy for amotivation and anhedonia.

8.4 Daytime Step Total Variation (DSTV)

Healthy daytime behavior is typically characterized by "jump processes"—discrete, purposeful bursts of movement.²⁹ In states of psychomotor retardation, these bursts flatten into "diffusion-like" movement: a sluggish, continuous, low-intensity state. By applying Total Variation mathematics—originally used in high-frequency finance to distinguish signal from noise—to step counts, we can quantify the loss of behavioral "burstiness." Lower total variation in 15-minute step bins across the wake window reflects the erosion of goal-directed activation, providing a mathematical quality score for daytime movement.

emergent biomarkers derived from the interaction of physical

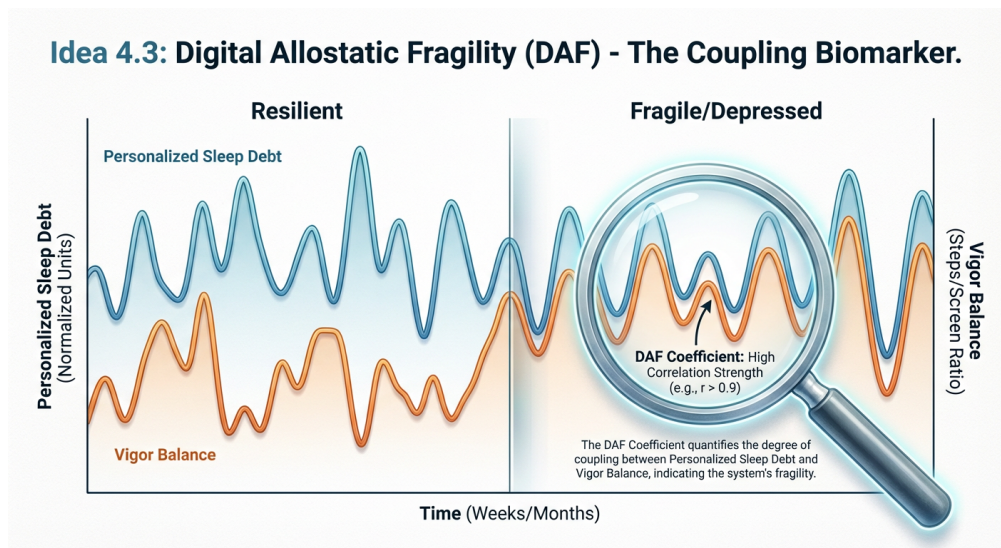


Figure 4: Conceptual diagram 4 from the biomarker discovery analysis.

9 Recommendation and Best Next Steps

Following a comprehensive evaluation of the proposed digital signatures against the Research Goal and Review Guidelines, the Asymmetric Behavioral Reactivity Profile (aBRP) is recommended as the primary candidate for initial implementation. Despite identified operational risks, the aBRP offers the most robust theoretical integration of the Effort-Based Decision Making framework and compositional data analysis, providing a scale-invariant measure of psychological fragility that is particularly well-suited for large, heterogeneous datasets.

While several ideas presented innovative approaches—specifically the Chrono-Digital Dissociation (CDD) Index for its treatment of missingness and the Digital Allostatic Fragility (DAF) for its focus on state-behavior coupling—the aBRP’s ability to normalize for baseline trait activity through intra-individual differences makes it a superior candidate for detecting the “reactive” components of depression severity.

Recommended Next Steps

Having completed an exhaustive literature search (2015–2025) and conceptual synthesis, the next steps focus on the empirical execution of the 7-step analysis workflow: 1. Pilot Data Transformation: Execute Step 2 of the workflow by calculating the Physical Activity-to-Screen Log-Ratio ($PASRL_t$) across the 144K observational rows. This must include the +1 Laplace smoothing constant to ensure mathematical stability. 2. Sensitivity Threshold Testing: Conduct a sensitivity analysis on the IQR versus Standard Deviation (SD) methods for defining sleep “stressors.” This will address the risk of “pathologizing normal variance” by testing which threshold (e.g., ± 1 SD or IQR) yields the strongest univariate signal. 3. Univariate Screening (Step 3): Perform Spearman ρ correlations with Benjamini-Hochberg FDR correction. Specifically, monitor for the “signal cancellation” effect in reactivity metrics to determine if bifurcating the reactivity by phenotype (Melancholic vs. Atypical) is required before univariate filtering. 4. Multivariate Validation (Step 4 & 5): Integrate the validated aBRP features into the Gradient Boosting Machine (GBM) and Random Forest models. Utilize SHAP (SHapley Additive exPlanations) importance analysis to quantify the incremental predictive value of the aBRP-Reactivity and aBRP-Variance features over static mean volume metrics. 5. Robustness Check for Data Density: Evaluate the impact of the 19-day average longitudinal window on the stability of the aBRP metrics. If aleatoric noise is too high, implement a minimum data threshold (e.g., ≥ 14 days) and assess the resulting “Survivor Bias” impact on the sample size. 6. Confounder Adjustment: Finalize the partial correlation models by adjusting for age, sex, and a “Weekday/Weekend” binary flag to ensure that the reactivity captured is psychogenic rather than a mere artifact of social jetlag.

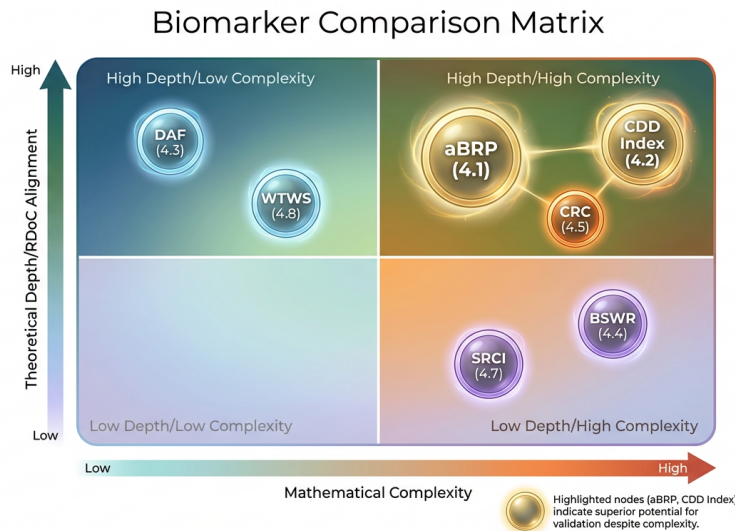


Figure 5: Conceptual diagram 5 from the biomarker discovery analysis.

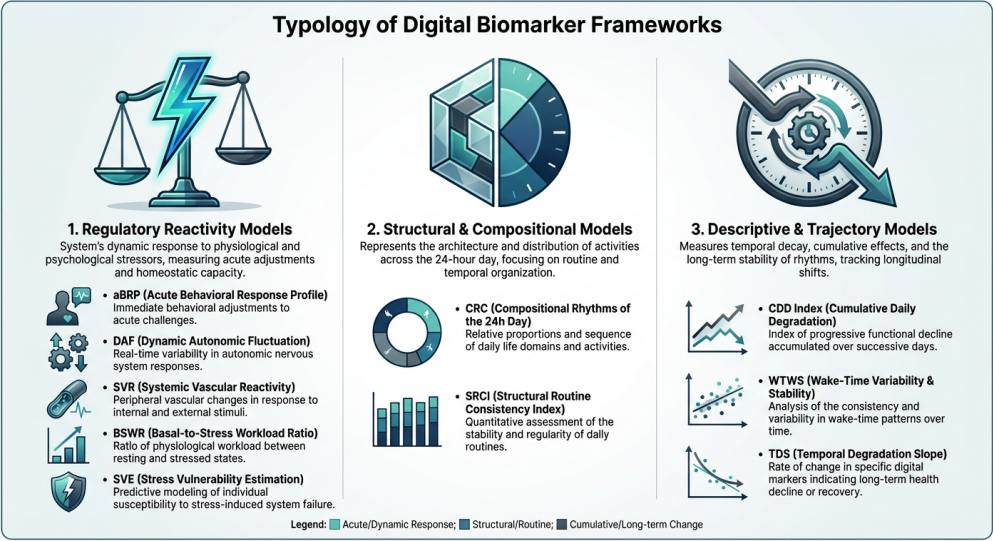


Figure 6: Conceptual diagram 6 from the biomarker discovery analysis.

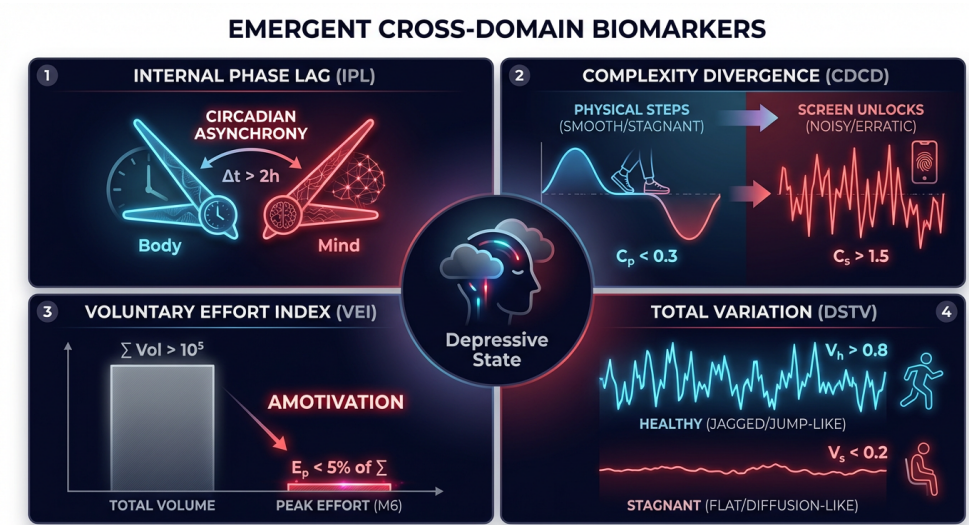


Figure 7: Conceptual diagram 7 from the biomarker discovery analysis.